THE BRITISH COLLEGE, THAPATHALI, KATHMANDU

Submitted By: Abhishek Chimariya

Orbund Id:11824

UWE Welcome Id: 25068602

Reflection:

OOP, Functions, and Modularity

Working on this group project was a valuable learning experience that helped me deepen my understanding of Object-Oriented Programming (OOP), functions, and modularity. Being responsible for one of the four divided parts of the project, I had the opportunity to focus specifically on implementing structured and reusable code while also coordinating with my teammates to ensure overall consistency.

One of the key aspects I learned through this project was the importance of modular design. By breaking down the program into smaller, manageable components, it became easier to develop, test, and debug individual sections without affecting the entire system. My part required me to design functions that were not only efficient but also reusable, which reinforced the concept that good programming is not just about making code work, but making it clean, scalable, and maintainable.

Additionally, applying Object-Oriented Programming concepts such as classes, objects, encapsulation, and abstraction gave me a clearer perspective on how large-scale programs are structured. I realized that OOP allows developers to model real-world scenarios effectively, making the code more intuitive and organized. While working on my section, I had to carefully think about how my classes and functions would interact with other parts of the project, which improved my ability to write compatible and collaborative code.

However, this project was not without challenges. One difficulty I faced was ensuring that my team integrated smoothly with the work of my teammates. Since each member was responsible for a different section, maintaining consistency in coding style, logic, and structure required continuous communication. At times, there were minor mismatches in how different parts were implemented, which led to debugging issues during integration. This taught me the importance of planning, clear communication, and setting common standards before starting a group project.

Another challenge was managing time effectively. While I understood the concepts theoretically, applying them practically required more time and effort than expected. This made me realize that practical implementation is essential for mastering programming concepts, and that time management plays a crucial role in project success.

Overall, this project significantly enhanced my technical skills as well as my teamwork abilities. It helped me understand how modularity improves code efficiency, how functions promote reusability, and how OOP provides a strong foundation for structured programming. More

importantly, it taught me the value of collaboration, communication, and adaptability in a group setting.

In conclusion, this experience has strengthened my confidence in programming and motivated me to further improve my skills. If given another opportunity, I would focus more on early planning and team coordination to make the development process smoother and more efficient.

Contribution:

Developed three role-based class modules Student, Teacher, and Exam extending a base User model to implement a structured academic management system. The student class enables students to manage their own profile, view grades and ECA records, and access a personal analytics dashboard. The Teacher class allows educators to look up any student's full profile, academic grades, and ECA data, along with analytics access. The Exam dept class provides administrative capabilities including updating student records and GPA entries (both semester and yearly), viewing complete student information, and generating analytics reports all backed by CSV-based data persistence and a shared analytics module.